

Data Preprocessing Report

1. Business Problem

The objective of this project is to prepare the Heart Failure Prediction Dataset for a machine-learning classification task. The business problem is to determine whether patient characteristics can be used to predict the presence of heart disease. The target variable is **HeartDisease**.

The purpose of preprocessing is to improve data quality and transform the raw dataset into a consistent numerical format suitable for machine-learning algorithms. The resulting model is intended for research and educational purposes and should not be considered a clinical diagnostic system.

2. Preprocessing Decisions

The dataset was first inspected for data types, missing values, duplicate records, potential outliers, and the distribution of the target variable. Numerical and categorical variables were handled separately because they have different characteristics.

The categorical variables identified were **Sex**, **ChestPainType**, **RestingECG**, **ExerciseAngina**, and **ST_Slope**. These variables were selected for one-hot encoding because their categories do not have a numerical order. One-hot encoding prevents the model from incorrectly interpreting category labels as having a meaningful ranking.

FastingBS was retained as a binary numerical variable because it already represents two possible states. The target variable, **HeartDisease**, was also retained separately as the classification outcome and was not scaled.

The continuous numerical variables considered for scaling were **Age**, **RestingBP**, **Cholesterol**, **MaxHR**, and **Oldpeak**. **StandardScaler** was selected because these variables have different numerical ranges.

Missing values and Duplicate records were assessed before preprocessing. The dataset had none.

3. Feature Engineering and Feature Relevance

The main feature transformations were one-hot encoding of categorical variables and standardization of continuous numerical variables. These transformations convert the original data into a numerical representation suitable for machine learning.

Feature relevance was investigated using correlation analysis, Random Forest feature importance, and mutual information. Correlation analysis was used to examine relationships between numerical variables and identify potentially redundant features. Random Forest feature importance was used to determine which variables contributed most strongly to predictions, while mutual information was used to identify features containing useful information about the target.

4. Outliers and Data Quality

Potential outliers were investigated using the Interquartile Range (IQR) method and boxplots. An observation can be considered a potential outlier when it falls below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$. Extreme observations were not automatically removed because the dataset contains clinical measurements. An unusually high or low value may

represent a legitimate patient characteristic rather than an error. Removing such observations without investigation could reduce the representativeness of the dataset and introduce bias.

Data types were also validated to ensure that numerical variables were stored as numerical values and categorical variables were correctly identified. These checks help ensure that encoding, scaling, and other preprocessing operations are applied correctly.

5. Justification of Methods and Machine-Learning Readiness

The selected preprocessing methods were based on the characteristics of the dataset. One-hot encoding was chosen because the categorical variables do not have a natural numerical order. StandardScaler was selected because the continuous variables have different ranges and scale-sensitive algorithms which can be affected by these differences. IQR provides a robust method for identifying potential outliers.

A preprocessing pipeline was used so that transformations are fitted using the training data only and then applied to the test data. This prevents data leakage and ensures that the evaluation of future models is more reliable.

After preprocessing, the dataset is suitable for subsequent machine-learning experimentation. The prepared data has appropriate data types, assessed duplicates and missing values, encoded categorical variables, appropriately transformed numerical variables, and a clearly separated target variable.

6. Key Observations and Conclusion

The preprocessing analysis showed that the dataset contains a mixture of numerical, binary, and categorical variables, requiring different preprocessing techniques. Categorical variables cannot be treated as ordinary continuous numbers because their category labels do not represent meaningful numerical distances. Numerical variables also have different ranges, making standardization appropriate for scale-sensitive algorithms.

Potential outliers were investigated rather than automatically removed because extreme clinical measurements may be valid observations. Feature relevance was assessed using multiple techniques to provide a broader understanding of which variables may contribute to predicting heart disease.

Overall, the dataset has been inspected, cleaned, transformed, and prepared for machine-learning experimentation. The preprocessing workflow included data-type validation, duplicate detection, missing-value assessment, outlier investigation, categorical encoding, numerical scaling, correlation analysis, and feature-importance analysis. The resulting dataset can be used for a subsequent binary classification project.

However, the dataset and any resulting machine-learning model should not be interpreted as a clinical diagnostic system. Real-world clinical use would require additional validation on representative patient populations, medical oversight, and evaluation of the model's safety, fairness, and reliability.